Role of Interprocess communication in performance of Distributed Systems

Badari Burli



Introductions

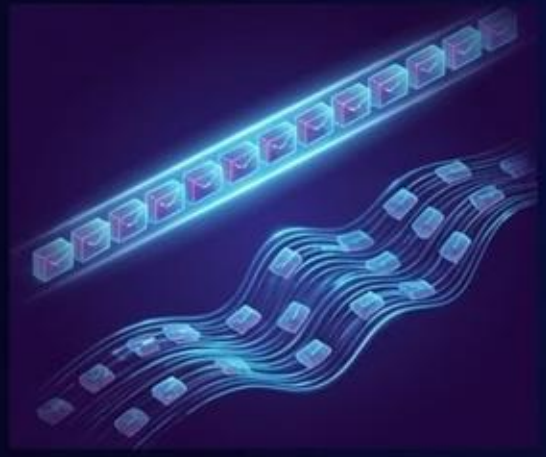


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Personal Blog	https://medium.com/@bburli
Suki Engineering Blog	https://medium.com/suki-engineering

Pillars of Distributed Systems

Consistency



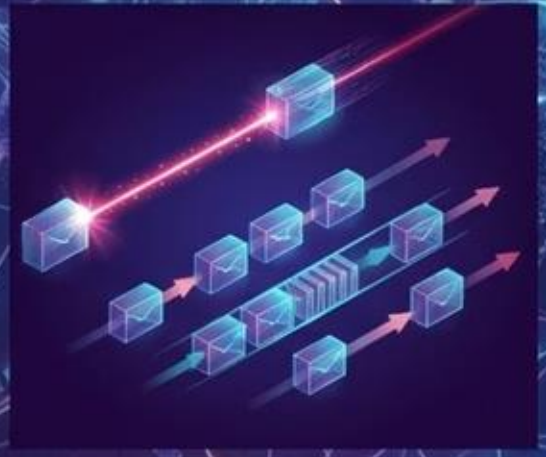
Strong at one end
Eventual consistency at another

Coordination



Orchestrated at one end to Choreographed at another

Communication

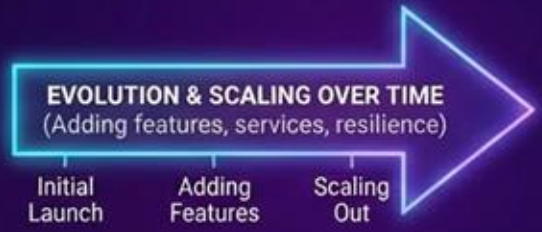


Synchronous at one end to Asynchronous at another

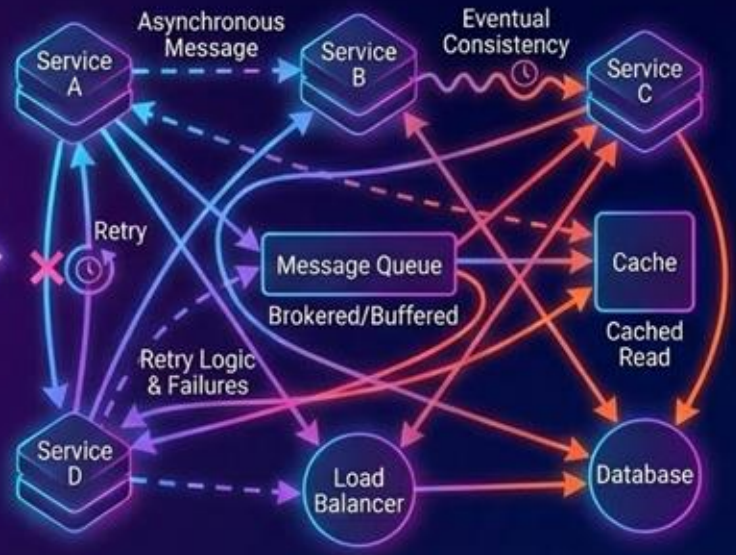
Evolution of Distributed System Communication: From Simple to Complex

DECEPTIVELY SIMPLE START





COMPLEX OVER TIME

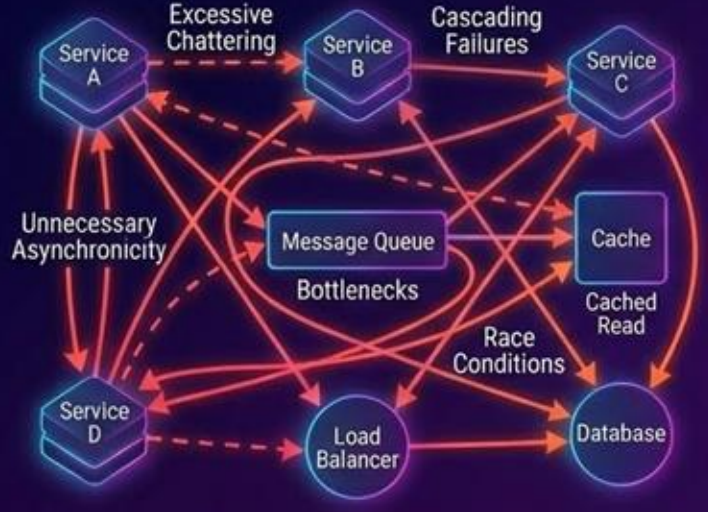


Seems easy to manage. Point-to-Point. Immediate response.

Hard to trace and debug. Asynchronous, Indirect paths, Distributed state, Latency & Failure modes.

Distributed Communication Complexity & Tradeoff Ignorance: Performance Issues

COMMUNICATION COMPLEXITY



Too many paths, opaque failures, mismanaged state

WRONG CHOICES & TRADEOFFS IGNORED

PERFORMANCE IMPACTS & TRADEOFFS

High Latency
Slows down user experience

Throttling & Backpressure
Overloads systems

Inconsistency & Stale Data
Data out of sync

Slow Response Times
Poor throughput

Ignoring consistency, availability, latency leads to bottlenecks

Ignoring tradeoffs leads to bottlenecks and unreliable systems.

THREE STRATEGIES

Lessons from real world of audio processing



BATTLE SCARS:

- Audio packets go through multiple hops until they're transcribed
- Had to handle failure cases everywhere!
- Stateful
- GPU hosting and Load Balancing challenges
- ^ Cost
- ^ Complexity





1. No communication - make it a modulith!

Don't introduce network if your domain problems need coupled communication.

BATTLE SCARS:

Redis Streams vs GCP Pub Sub vs Kafka:

- Durability
- Latency
- Recoverability

gRPC:

- Extremely fast, Schema driven
- Binary format; Not for external customers
- Poor browser compatibility for bi-directional audio handling



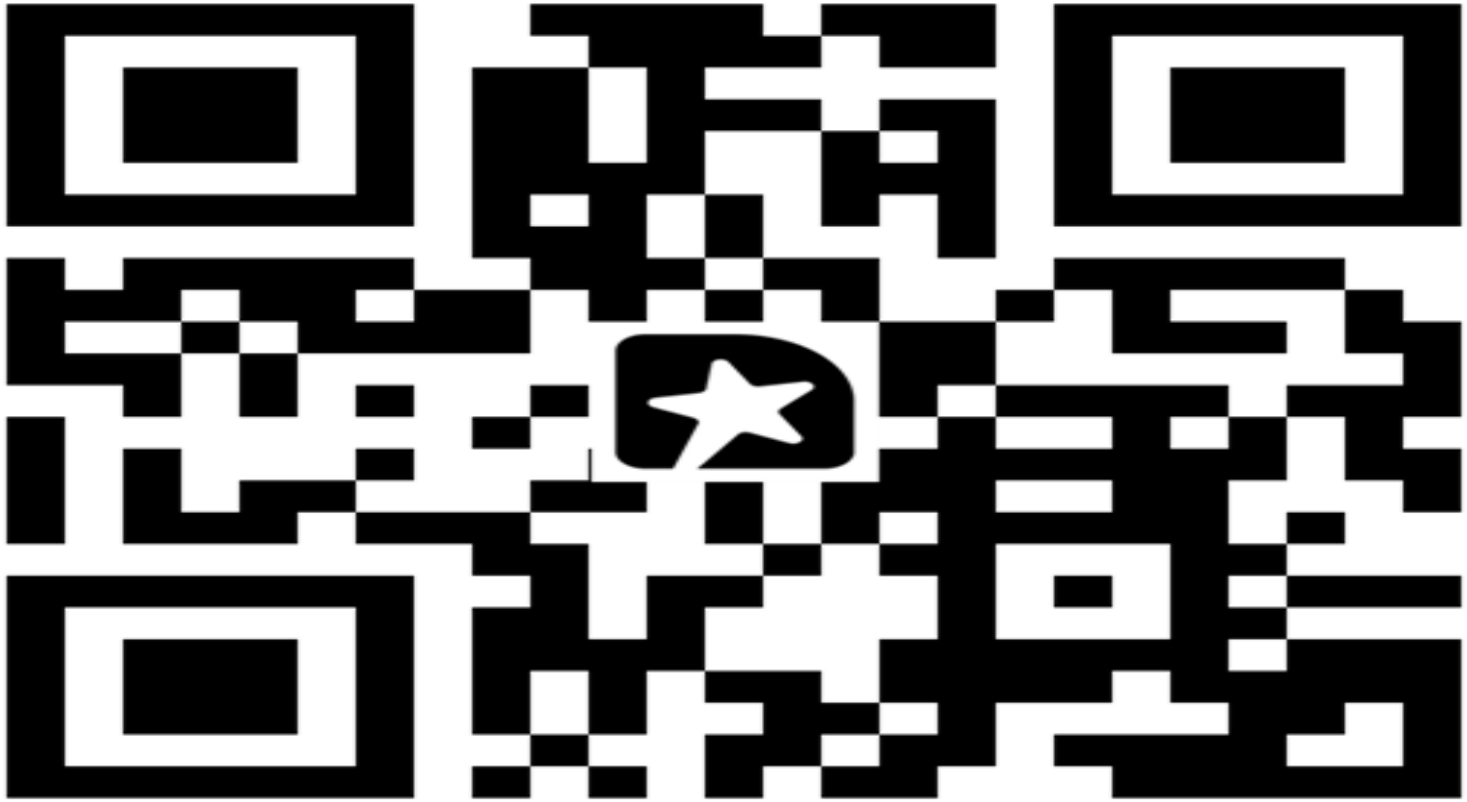
2. **Use right method**: external = http/rest, internal = gRPC, decoupled and **durable** = pub/sub, real time / ephemeral = redis streams, client streaming, server streaming and bi-directional (remember hotspot problems)

BATTLE SCARS:

Start simple:



3. Don't force technologies to dictate how you communicate. Choose yourself, choose wisely and simply.



Role of interprocess communication technologies in performance of a distributed system